



BASIC MULTIDISCIPLINARY PROJECT

Identification of Systemic Lupus Erythematosus (Lupus) Using Machine Learning Algorithms

*Submitted in partial fulfillment of requirements for the
award of*

Bachelor of Technology (B. Tech)

Academic year : 2025-2026

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CONTENTS OF THE PROJECT

- ❖ **ABSTRACT**
- ❖ **INTRODUCTION**
- ❖ **OBJECTIVES**
- ❖ **METHODOLOGY**
- ❖ **RANDOM FOREST ALGORITHM**
- ❖ **WORKING PRINCIPLE**
- ❖ **SOFTWARE DESCRIPTION AND TOOLS USED**
- ❖ **SYSTEM ARCHITECTURE**
- ❖ **ALGORITHM AND FLOWCHART**
- ❖ **RESULTS AND OBSERVATION**
- ❖ **FUTURE SCOPE AND ENHANCEMENTS**
- ❖ **CONCLUSION**
- ❖ **REFERENCES**

1. Abstract

This project focuses on developing an Artificial Intelligence-based Lupus Disease Prediction System using Machine Learning techniques. Lupus is a complex autoimmune disorder that is difficult to diagnose at an early stage due to overlapping symptoms with other diseases. In this project, a medical dataset related to autoimmune disorders was obtained from Kaggle and processed using Python in the Google Colab environment. Important clinical features such as Age, ANA test results, Anti-dsDNA levels, CRP, ESR, fever, joint pain, skin rashes, and fatigue were selected as input parameters. The dataset was cleaned, filtered, and prepared to identify Lupus and non-Lupus cases. A Random Forest Classifier algorithm was used to train the model because of its high accuracy and ability to handle complex medical data.

The trained model was further enhanced with feature importance visualization to understand the influence of each clinical factor on the prediction results. An interactive user interface was also developed using widgets to allow users to input patient details easily and obtain real-time Lupus risk predictions. The system calculates the probability of Lupus occurrence and categorizes the patient into Low Risk, Moderate Risk, or High-Risk levels based on predefined thresholds. This project demonstrates how Machine Learning can support early disease prediction and assist healthcare professionals in making faster and more informed clinical decisions.

2. Introduction

Systemic Lupus Erythematosus (SLE), commonly known as Lupus, is a chronic autoimmune disease in which the immune system mistakenly attacks healthy tissues and organs. This condition can affect different parts of the body such as the skin, joints, kidneys, and nervous system, leading to inflammation and long-term complications. One of the major challenges in diagnosing lupus is that its symptoms often overlap with those of other autoimmune diseases, making early detection difficult. Traditional diagnostic methods rely on clinical observation, laboratory tests, and physician expertise, which can sometimes be time-consuming and prone to delays in accurate diagnosis. Therefore, there is a growing need for intelligent systems that can assist healthcare professionals in identifying lupus at an early stage.

With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), it has become possible to develop predictive systems that can analyze medical data and detect patterns related to diseases. In this project, a machine learning-based system is developed to predict the probability of lupus using patient clinical data such as age, laboratory test results, and symptoms. A Random Forest algorithm is used due to its high accuracy and ability to handle complex medical datasets. The system also includes feature visualization and an interactive user interface that allows users to input patient information and receive real-time risk predictions. This project demonstrates how machine learning can support early diagnosis, improve decision-making, and contribute to more efficient healthcare solutions.

3. Objectives

The main objective of this project is to develop a machine learning-based system capable of predicting the probability of Systemic Lupus Erythematosus (SLE) using patient clinical data. The system aims to assist in early detection of lupus by analyzing important medical features such as laboratory test results and common symptoms. By using machine learning techniques, the project seeks to improve prediction accuracy and reduce the time required for diagnosis compared to traditional methods.

The specific objectives of this project are:

1. To collect and preprocess a medical dataset related to autoimmune disorders obtained from public sources such as Kaggle.
2. To select relevant clinical features such as Age, ANA, Anti-dsDNA, CRP, ESR, fever, joint pain, skin rash, and fatigue for training the model.
3. To develop and train a Random Forest machine learning model for predicting lupus risk based on patient data.
4. To visualize feature importance in order to identify the most significant factors influencing lupus prediction.
5. To design an interactive user interface that allows users to input patient details and receive real-time risk predictions.
6. To evaluate the performance of the model using test data and analyze its prediction accuracy.
7. To demonstrate the usefulness of machine learning techniques in supporting healthcare professionals in clinical decision-making.

4. Methodology

The methodology of this project involves developing a machine learning-based system to predict Lupus disease using clinical data. The process includes data collection, preprocessing, model training, visualization, and prediction through an interactive interface.

3.1 Data Collection

- The dataset was obtained from the Kaggle platform, containing autoimmune disorder records.
- It includes clinical features such as Age, ANA, Anti-dsDNA, CRP, ESR, and symptoms.
- The dataset was uploaded and processed using Python in Google Colab.

3.2 Data Preprocessing

- Column names were cleaned, and relevant disease records were filtered.
- A target column was created (Lupus = 1, Non-Lupus = 0).
- Missing values were handled and clinical calibration was applied to improve data quality.

3.3 Feature Selection

- Important clinical features related to Lupus diagnosis were selected.
- Selected inputs include Age, ANA, Anti-dsDNA, CRP, ESR, and symptom indicators.

- These features were used as input variables for model training.

3.4 Model Training Using Random Forest

- The dataset was divided into training (80%) and testing (20%) data.
- A Random Forest Classifier was trained using multiple decision trees.
- The final prediction was obtained using majority voting among trees.

3.5 Feature Importance Visualization

- Feature importance values were extracted from the trained model.
- A bar chart was generated using Matplotlib and Seaborn.
- The graph highlights the most important clinical features.

3.6 Development of Interactive User Interface

- An interactive interface was created using Python widgets.
- Sliders, dropdowns, and checkboxes were used for patient input.
- A prediction button was added to generate results.

3.7 Prediction and Risk Assessment

- User input data is converted into model-compatible format.
- The model predicts Lupus probability using `predict_proba`.
- The output is displayed as Low, Moderate, or High Risk with probability percentage.

5. Random Forest Algorithm

Random Forest is a supervised machine learning algorithm used for classification and prediction tasks. It combines multiple decision trees to produce accurate and reliable results. In this project, the Random Forest algorithm is used to predict the probability of Systemic Lupus Erythematosus (SLE) using patient clinical data such as laboratory test results and symptoms.

5.1 Why Random Forest is Used?

- Random Forest provides higher accuracy compared to algorithms like Decision Trees and Logistic Regression.
- A single Decision Tree may give unstable results, but Random Forest improves stability by using multiple decision trees.
- It can handle nonlinear relationships between medical features better than Logistic Regression.
- It reduces the problem of overfitting, making predictions more reliable.
- It performs well even when the dataset contains many features and complex medical patterns.
- It is suitable for medical prediction systems, where reliability and accuracy are very important.

5.2 Working Principle of Random Forest

- The dataset is divided into multiple random subsets using a method called bootstrapping.
- For each subset, a decision tree is created using randomly selected features.

- Each decision tree independently predicts whether the patient has lupus or not.
- All trees generate their own predictions.
- The final prediction is determined using a majority voting mechanism among all trees.
- Combining multiple trees improves the accuracy and reliability of the model.

5.3 Advantages of Random Forest

- Provides high prediction accuracy compared to many other algorithms.
- Reduces overfitting by combining multiple decision trees.
- Can handle large datasets efficiently.
- Works well even when some data values are missing.
- Provides feature importance, helping identify the most important clinical factors affecting lupus prediction.
- Suitable for complex medical datasets with multiple symptoms and laboratory test values.
- Produces stable and consistent results even with different datasets.

6. Working Principle

The working principle of the Lupus Disease Prediction System describes how the system processes patient input data and generates prediction results using the trained machine learning model.

The working process of the system is as follows:

1. User Input

- The user enters patient details such as age, test results, and symptoms using the interactive interface.

2. Data Processing

- The input data is converted into structured format compatible with the trained Random Forest model.

3. Prediction Generation

- The trained model analyzes the input data and calculates the probability of lupus occurrence.

4. Risk Classification

- Based on the predicted probability, the system classifies the patient into Low Risk, Moderate Risk, or High Risk categories.

5. Output Display

- The system displays the prediction result along with probability percentage and visual risk indicator.

7. Software Description and Tools used

This section describes the software platforms, programming languages, and libraries used to develop the Lupus Disease Prediction System. These tools were selected because they provide efficient data processing, machine learning implementation, and interactive visualization capabilities.

6.1 Software Description

The software developed in this project is an AI-based diagnostic support system designed to predict the probability of Systemic Lupus Erythematosus (SLE) using patient clinical data. The system integrates data preprocessing, machine learning model training, visualization, and an interactive user interface into a single workflow.

Key features of the developed software include

- Accepts patient clinical inputs such as age, laboratory test results, and symptoms.
- Processes medical data using preprocessing techniques such as filtering and missing value handling.
- Uses a trained Random Forest machine learning model to predict the probability of lupus.
- Displays prediction results in the form of probability percentages and risk categories.
- Provides visual outputs such as feature importance graphs and risk gauge indicators.
- Includes an interactive user interface that allows easy input of patient data.
- Supports real-time prediction and analysis using trained machine learning models.

4.2 Tools and Technologies Used

The following tools and technologies were used in the development of the Lupus Disease Prediction System:

- **Google Colab:** Used as the cloud-based platform to write and execute Python machine learning code.
- **Python:** Used as the main programming language to develop the machine learning model and interface.
- **Pandas:** Used to load, clean, and manage the dataset in tabular format.
- **NumPy:** Used to perform numerical operations and handle array-based calculations.
- **Scikit-learn:** Used to implement the Random Forest algorithm and perform model training and prediction.
- **Matplotlib:** Used to create graphical visualizations such as risk bars and plots.
- **Seaborn:** Used to generate feature importance graphs with improved visualization style.
- **IPyWidgets:** Used to create the interactive user interface with sliders, dropdowns, and buttons.
- **Kaggle Dataset:** Used as the source of medical data for training and testing the machine learning model.

8. System Architecture

The System Architecture of the Lupus Disease Prediction System represents the overall structure and workflow of the developed model. The system is designed in a modular manner, where each module performs a specific function such as data processing, model training, and prediction generation. The architecture ensures smooth data flow from input to output, enabling efficient and accurate prediction of lupus risk.

The system consists of the following main components:

1. Dataset Module

- This module is responsible for loading the medical dataset obtained from Kaggle.
- The dataset contains patient clinical features such as age, laboratory test results, and symptoms.
- The data is stored in CSV format and loaded into the system using Python libraries.

2. Data Preprocessing Module

- This module cleans and prepares the dataset for model training.
- It removes unwanted spaces from column names and filters relevant disease records.
- Missing values are handled, and a target column is created to classify lupus and non-lupus cases.

- Clinical calibration techniques are applied to improve dataset realism.

3. Machine Learning Model Module

- This module trains the Random Forest algorithm using processed dataset features.
- The dataset is divided into training and testing subsets.
- The trained model learns patterns from medical data and predicts lupus probability.
- Feature importance analysis is performed to identify significant clinical factors.

4. User Interface Module

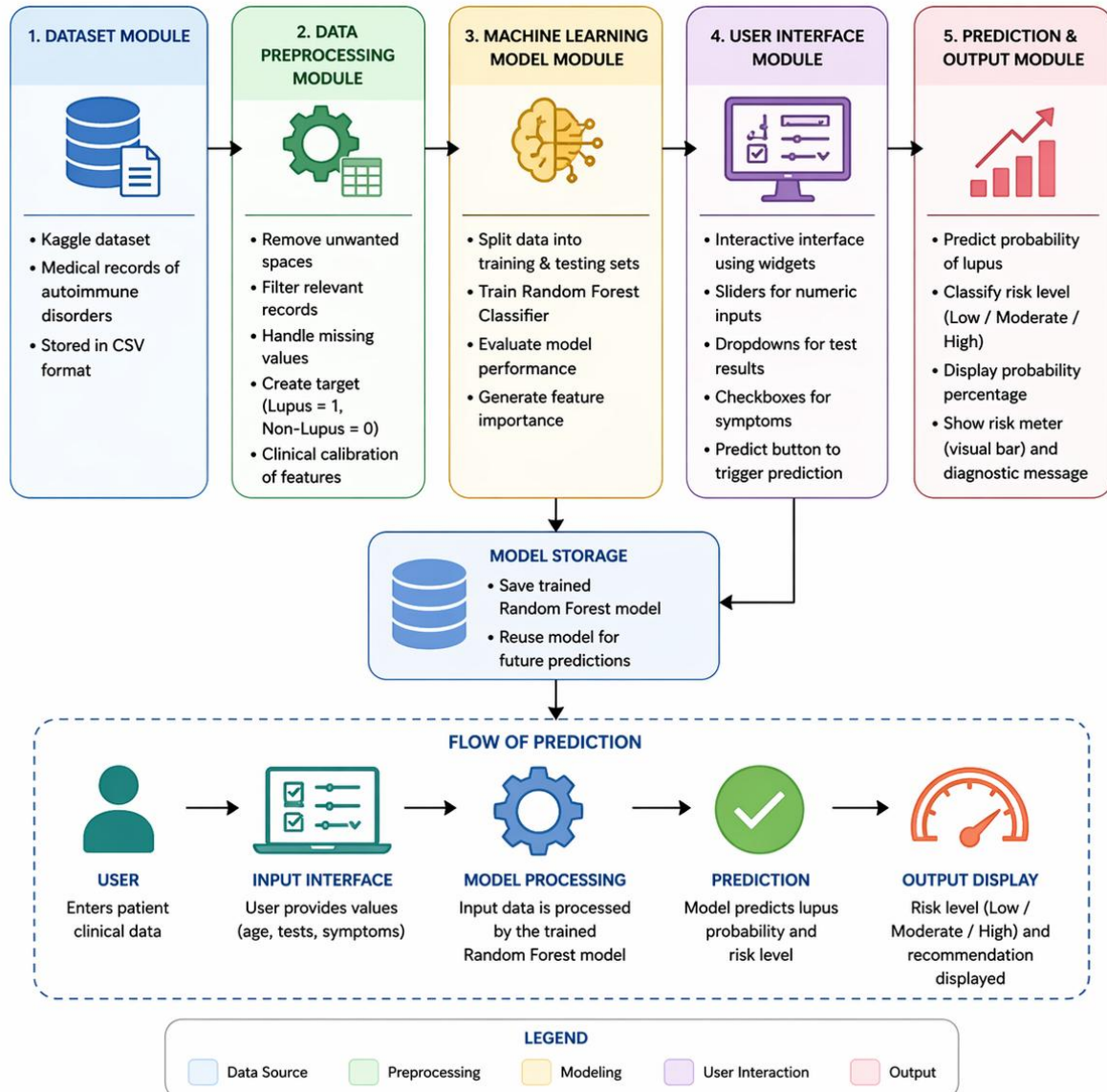
- This module provides an interactive interface using sliders, dropdown menus, and checkboxes.
- Users can input patient clinical information easily.
- A prediction button is used to trigger the lupus risk prediction process.

5. Prediction and Output Module

- This module processes user inputs and generates lupus risk probability.
- The system displays results as probability percentage and risk category.
- A visual risk bar is generated to indicate Low, Moderate, or High risk levels.

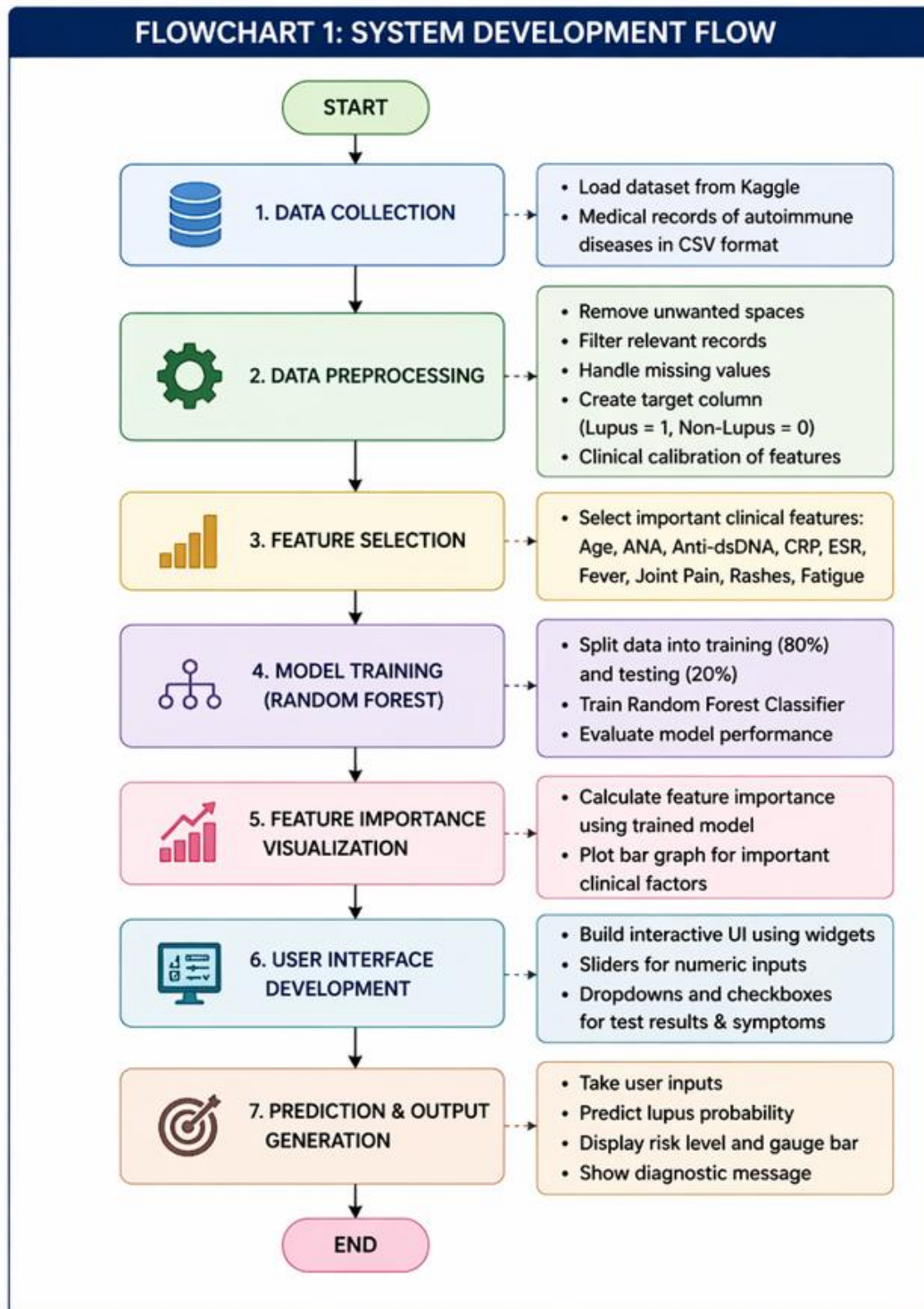
SYSTEM ARCHITECTURE

Lupus Disease Prediction System Using Machine Learning

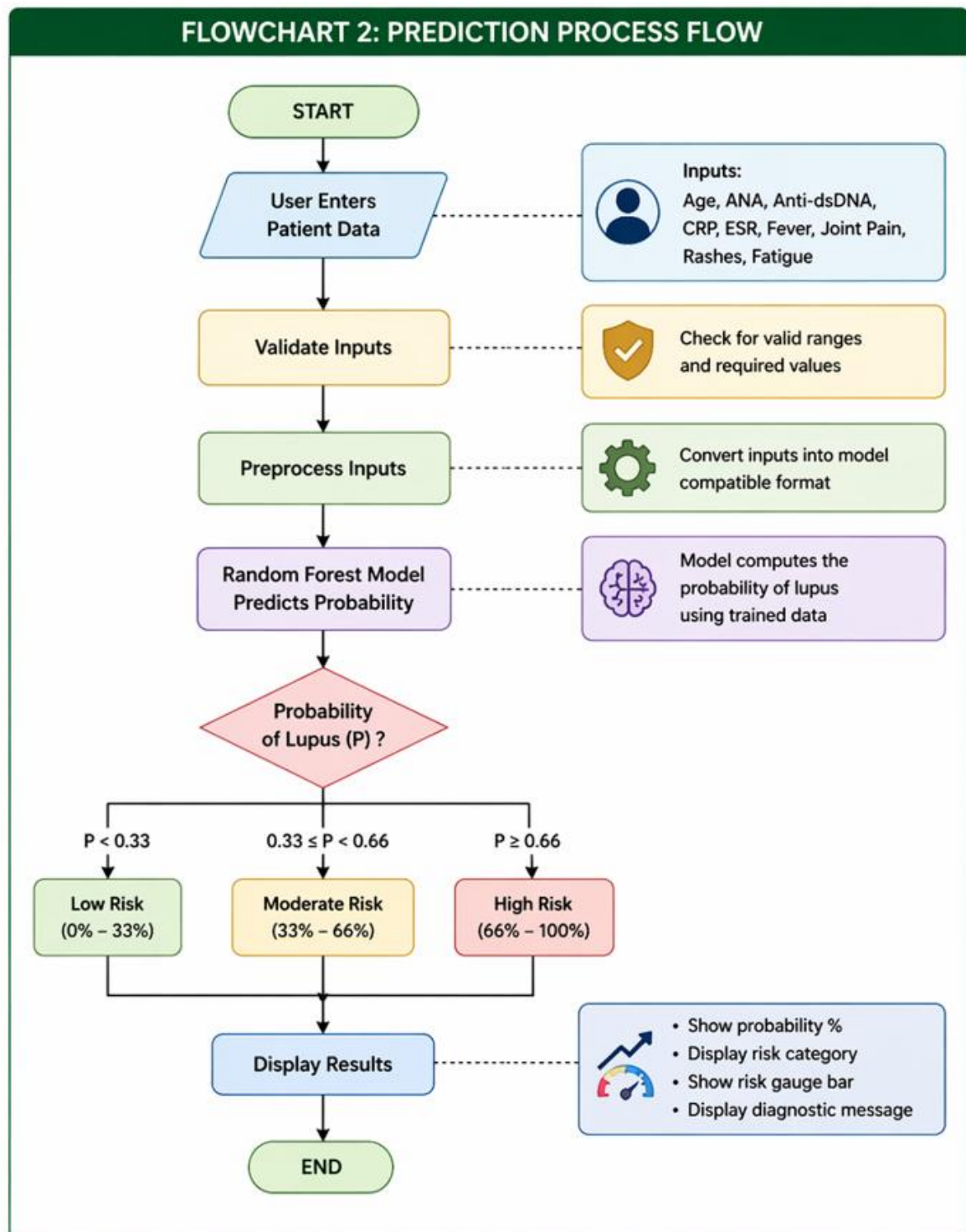


9. Algorithm and Flowchart

9.1 System Development Flow



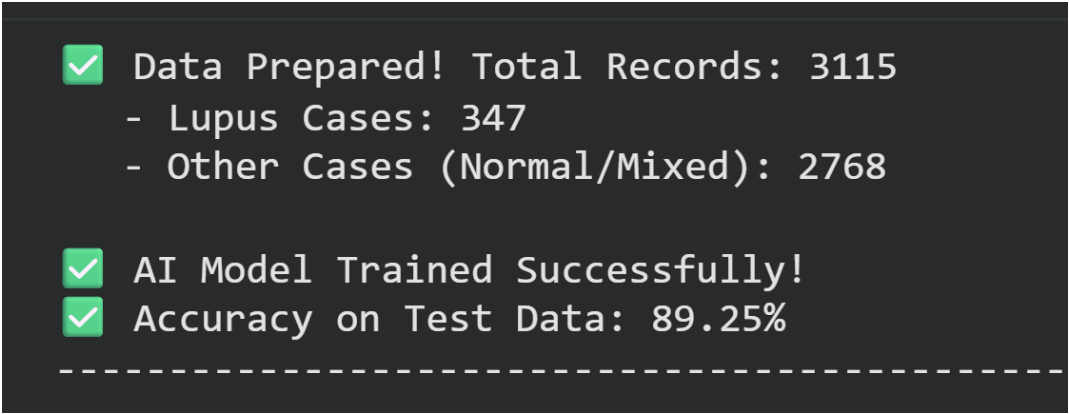
9.2 Prediction Process Flow



10. Results and Observation

The results of the Lupus Disease Prediction System were evaluated using the trained Random Forest machine learning model. After preprocessing the dataset and selecting relevant clinical features, the dataset contained a total of 3115 records, including 347 lupus cases and 2768 non-lupus cases. The dataset was divided into training and testing subsets to evaluate the model performance effectively.

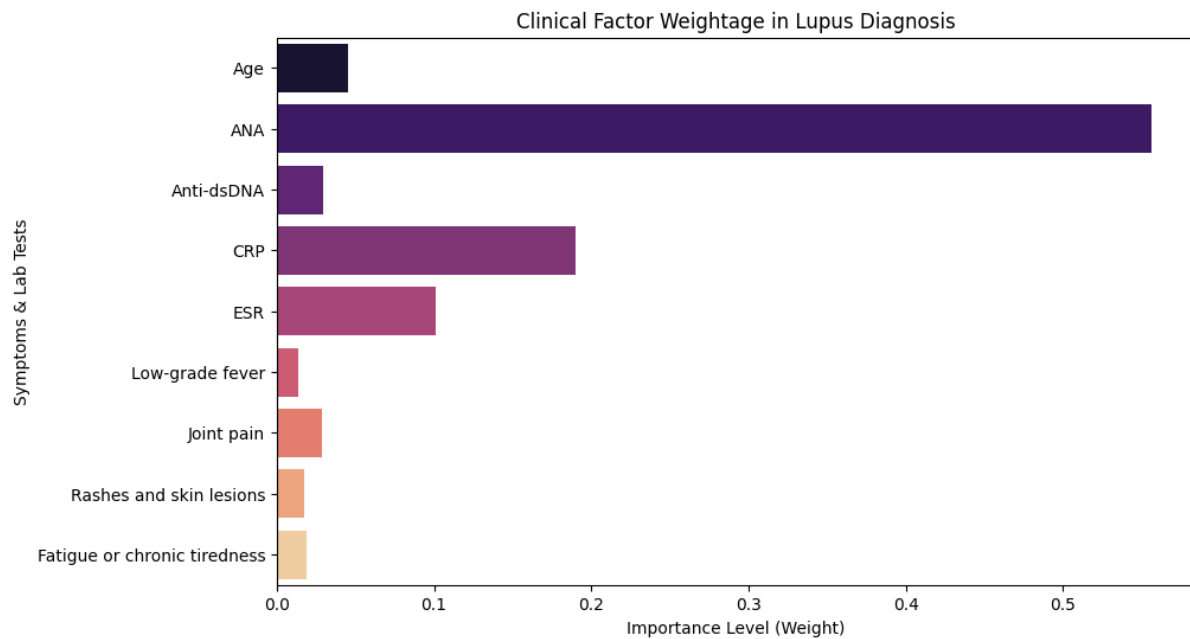
The Random Forest model was successfully trained using the training dataset, and its performance was evaluated using the testing dataset. The model achieved an accuracy of 89.25%, indicating that it correctly predicted lupus cases in most instances. This level of accuracy demonstrates that the selected features and machine learning approach are effective in identifying patterns associated with lupus disease.

A dark-themed terminal window with green checkmarks and white text. It displays the completion of data preparation and model training, along with the final accuracy.

```
✓ Data Prepared! Total Records: 3115
  - Lupus Cases: 347
  - Other Cases (Normal/Mixed): 2768

✓ AI Model Trained Successfully!
✓ Accuracy on Test Data: 89.25%
-----
```

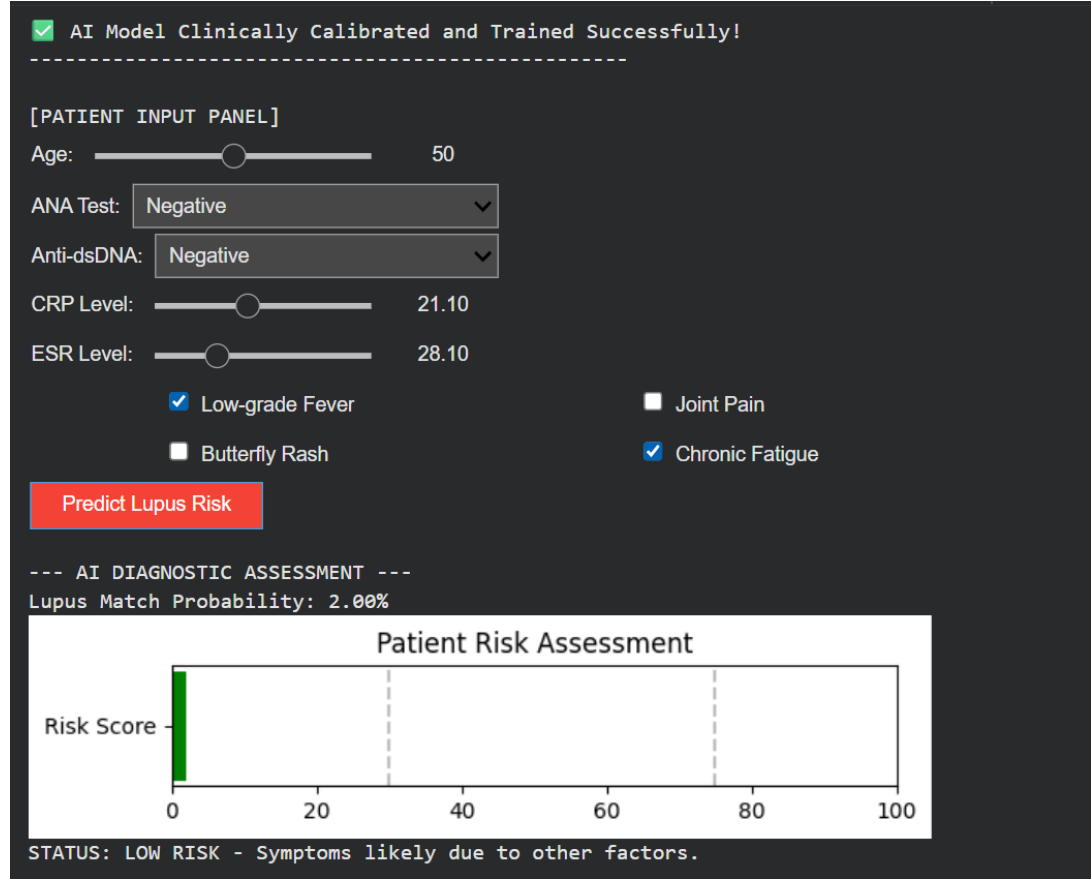
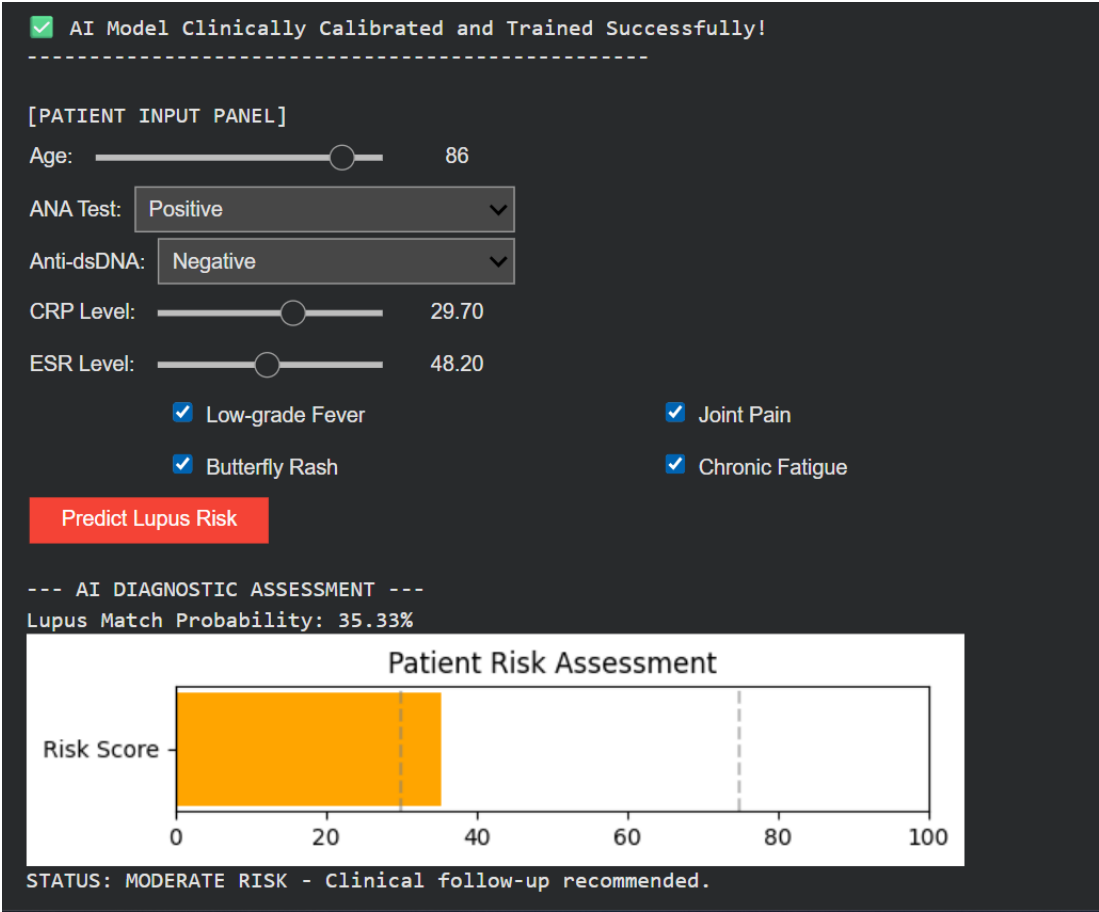
Feature importance analysis was performed to determine the contribution of each clinical parameter to the prediction results. The generated bar graph showed that ANA test results had the highest importance, followed by CRP and ESR levels, indicating their strong influence in lupus prediction. This observation supports the relevance of these clinical features in medical diagnosis.



(Feature Importance Graph showing clinical factor weightage in Lupus diagnosis.)

The bar graph shows that the ANA test has the highest importance among all clinical features, indicating that it plays a major role in lupus prediction. Other features such as CRP and ESR levels also show significant contribution, while symptoms such as fatigue, fever, and rashes have relatively lower importance. This result confirms that laboratory test parameters are highly influential in identifying lupus cases.

An interactive user interface was developed to allow users to input patient data and obtain real-time predictions. The system successfully generated lupus probability values and classified patients into Low Risk, Moderate Risk, and High Risk categories. A visual risk indicator was displayed to improve result interpretation. The successful implementation of prediction and visualization features demonstrates the practical usability of the developed system in assisting early diagnosis.



11. Future Scope and enhancements

The developed Lupus Disease Prediction System can be further improved and expanded in several ways to enhance its performance and practical usability. The following future improvements can be considered:

- **Use of Larger Datasets:** The model accuracy can be improved by training with larger and more diverse medical datasets collected from hospitals and research centers.
- **Integration with Hospital Systems:** The system can be integrated with Electronic Health Records (EHR) to automatically fetch patient data and provide real-time predictions.
- **Development of Mobile or Web Application:** The system can be converted into a mobile or web-based application to make it accessible to doctors and patients from remote locations.
- **Use of Advanced Machine Learning Models:** More advanced algorithms such as Deep Learning or Neural Networks can be used to improve prediction accuracy and performance.
- **Continuous Model Updating:** The system can be designed to update automatically with new patient data to improve prediction reliability over time.
- **Addition of More Clinical Features:** Including additional medical parameters such as genetic data, imaging results, or medical history can enhance prediction capability.

12. Conclusion

The Lupus Disease Prediction System developed in this project demonstrates the effective use of machine learning techniques for assisting in medical diagnosis. The system was designed to predict the probability of Systemic Lupus Erythematosus (SLE) using important clinical features such as Age, ANA, Anti-dsDNA, CRP, ESR, fever, joint pain, rashes, and fatigue. The dataset obtained from the Kaggle platform was carefully preprocessed and filtered to improve prediction reliability. The Random Forest algorithm was selected as the main machine learning model due to its ability to handle complex medical datasets and produce accurate predictions.

The trained model achieved an accuracy of 89.25%, indicating reliable performance in identifying lupus-related patterns from patient data. Feature importance analysis showed that laboratory test parameters such as ANA, CRP, and ESR have a major influence on prediction results. In addition, visualization techniques were used to represent the importance of different clinical features, improving the interpretability of the model. An interactive user interface was also developed to allow users to input patient data easily and obtain real-time risk predictions.

Overall, this project demonstrates the potential of machine learning technology in supporting early disease detection and improving decision-making in healthcare applications. The developed system successfully integrates data preprocessing, model training, visualization, and prediction into a single workflow. With further improvements such as larger datasets and advanced algorithms, this system can be expanded into a more practical diagnostic tool that assists healthcare professionals in detecting lupus cases more efficiently and accurately.

13. References

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